QML Lab • research project 06

Quantum Autoencoder

Compressing quantum states into fewer qubits and recovering them.

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Quadrant	QQ (quantum data, quantum processor)	**Track**	Variational
Difficulty	Medium	**Tools**	PennyLane
Builds on	Modules 2 and 5 of the QML Lab	**Workspace**	`qml_projects/quantum-autoencoder/`

Summary. A quantum autoencoder learns a unitary that squeezes a family of quantum states into fewer qubits, so that the discarded qubits always end up in a fixed reference state. Because the inputs and the processing are both quantum, this is a QQ project. It studies which state families compress, how far, and how robust the trained encoder is to noise.

1 Background and key concepts

1.1 Compression of quantum data

Let $\{|\psi_i\rangle\}$ be a family of n-qubit states, for example ground states of a Hamiltonian at different couplings. If all of them lie in a subspace of dimension at most 2^{n_A}, a unitary U can map them onto n_A "latent" qubits, leaving $n_B = n - n_A$ "trash" qubits in $|0\rangle$. Decoding applies $U^\dagger$ after resetting the trash qubits (Romero et al., 2017).

1.2 Training objective

For a parameterised encoder $U(\theta)$, the trash fidelity of state ψ_i is

$$F_i(\theta) = \mathrm{Tr}\Big[\big(I_A \otimes |0\rangle\langle 0|_B\big)\, U(\theta)\, |\psi_i\rangle\langle\psi_i|\, U(\theta)^\dagger \Big], \tag{1}$$

and training minimises $C(\theta) = 1 - \frac{1}{N}\sum_i F_i(\theta)$. If $C = 0$, decoding reproduces every state exactly. On hardware, overlaps such as this are often measured with the swap test (Buhrman et al., 2001), which returns 0 with probability $\frac{1}{2}\big(1 + |\langle a|b\rangle|^2\big)$. Pepper et al. (2019) realised a quantum autoencoder experimentally with photonic qutrits.

1.3 Why this is QQ

The encoder acts on quantum states directly, without measuring them. In simulation the input states are prepared classically, which hides the real use case: compressing the output of a quantum simulation or a quantum sensor before storing or transmitting it.

2 Research questions

The project is open-ended: these questions are the starting point, not the finish line. Add new ones to the README and the journal as they come up.

RQ1 Which state families compress well (Ising ground states, GHZ-like, W-like, Haar-random)?

RQ2 How does reconstruction fidelity fall when compressing further (3 to 2 versus 3 to 1 qubits)?

RQ3 How sensitive is a trained encoder to depolarizing noise?

3 Method

3.1 Data

State families on 3–4 qubits: ground states of the transverse-field Ising model for a range of field strengths (shared with Project 12), GHZ states with varying relative phase, W-like states, and Haar-random states as a negative control. Separate training and test states from each family.

3.2 Models

Hardware-efficient encoder with L layers of R_Y / R_Z rotations and CNOT rings on all qubits; cost from exact trash-qubit probabilities, and later from finite-shot swap tests; Adam optimiser.

3.3 Baselines

Classical reference: the rank of the state family's density matrix, which bounds the best possible compression; Haar-random states, which should not compress.

3.4 Metrics

- Trash fidelity and reconstruction fidelity on test states.
- Compression ratio n_A / n.
- Fidelity under depolarizing noise p.

4 Planned experiments

Each experiment becomes one `[[experiments]]` entry in the workspace, so its runs, parameters and metrics are stored and compared in the portal.

ID	Experiment	What varies	Main metrics
E1	3 to 2 qubits	Ising family	test reconstruction fidelity
E2	3 to 1 qubit	Ising and GHZ families	fidelity drop
E3	Negative control	Haar-random states	fidelity vs rank bound
E4	Noise	depolarizing $p \in [0, 0.05]$	fidelity
E5	Swap-test cost	shots M	training stability

Table 1: Experiment plan. Results go into the journal and the results log (section 9).

5 Running it in the QML Lab

The workspace lives in `qml_projects/quantum-autoencoder/`. Scripts import three helpers from `qml_lab/plots.py`:

- `params()` reads the parameters from the portal's run form (defaults when run by hand);
- `save()` stores figures with the run;
- `log_metrics()` records numbers that appear in the run table.

```
qml_projects/quantum-autoencoder/
  project.toml          metadata + experiments
  README.md             goal, open questions, ideas
  experiments/          runnable scripts
  docs/main.tex         this document
```

First experiment: This experiment is *planned*: write the script, add the block below to `project.toml`, then run `python manage.py sync_projects`.

```
[[experiments]]
slug = "compress-ising"
title = "Compress Ising ground states from 3 to 2 qubits"
script = "experiments/e01_compress_ising.py"
figure = "fidelity.png"

[experiments.params]
n_qubits = 3
n_trash = 1
n_layers = 3
steps = 200
family = "ising"
noise = 0.0
seed = 0
```

```
# from the portal folder: pick up changes to project.toml
python manage.py sync_projects
cd qml_projects/quantum-autoencoder
# run by hand with the default parameters
PYTHONPATH=../../qml_lab python experiments/e01_compress_ising.py
```

6 Quality bar

- A classical baseline with a similar number of trainable parameters, tuned with the same care (Bowles et al., 2024).
- At least five random seeds per configuration, reported as mean $\pm$ standard deviation.
- How results change with the number of qubits.
- At least one run under a realistic noise model.
- Always compare achieved fidelity with the rank bound of the state family.

7 Open directions

- Denoising autoencoders that remove noise from quantum states.
- Compress Ising states before feeding them to a QCNN (Project 12).
- Run a small encoder on hardware with a swap-test cost.

8 Risks and limitations

- Simulation prepares inputs classically, so the practical advantage of **QQ** processing is not visible.
- Deeper encoders on more qubits may hit barren plateaus.

9 Results log

Summarise findings here as they become stable; day-to-day notes belong in the portal journal.

Date	Run	Finding	Next step

References

Bowles, J., Ahmed, S. and Schuld, M. (2024) 'Better than classical? The subtle art of benchmarking quantum machine learning models', Preprint, arXiv:2403.07059. Available at: https://arxiv.org/abs/2403.07059.

Buhrman, H., Cleve, R., Watrous, J. and de Wolf, R. (2001) 'Quantum fingerprinting', *Physical Review Letters*, 87(16), 167902. Available at: https://doi.org/10.1103/PhysRevLett.87.167902.

Pepper, A., Tischler, N. and Pryde, G. J. (2019) 'Experimental realization of a quantum autoencoder: The compression of qutrits via machine learning', *Physical Review Letters*, 122(6), 060501. Available at: https://doi.org/10.1103/PhysRevLett.122.060501.

Romero, J., Olson, J. P. and Aspuru-Guzik, A. (2017) 'Quantum autoencoders for efficient compression of quantum data', *Quantum Science and Technology*, 2(4), 045001. Available at: https://doi.org/10.1088/2058-9565/aa8072.